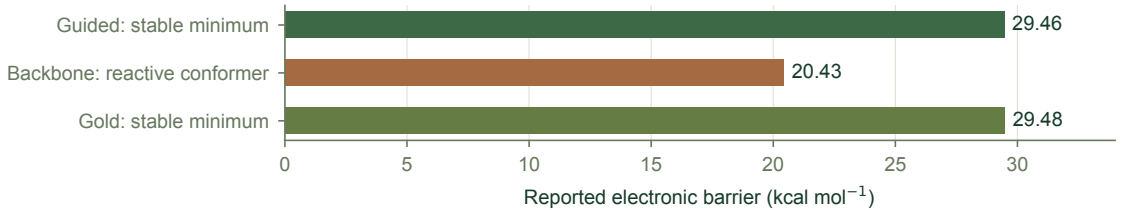
a **HA8: the reference state changes the barrier**



b **HH1: the same barrier, different evidence closure**



c **NU6: an energy difference needs a pathway identity**

Concerted TS −247.030042392 Ha

Betaine INT −247.060060699 Ha

Difference = 18.836772 kcal mol^{-1}

Guided: concerted saddle region; target INT → P saddle unresolved

Backbone: step-2 closure = 18.837